

Adaptive optimization-based bound tightening with finite certificates

October 2026

Abstract

Optimization-based bound tightening can reduce a search tree while increasing total solving time. This report develops finite checks that separate possible tightening benefit from computational cost. Feasible points bound the benefit of a frozen relaxation round. A finite pool of points feasible in the relaxation rebuilt on its own coordinate hull protects an inner box through every subsequent round of the same monotone relaxation family, with explicit cutoff and domain reuse conditions. A uniform matrix bound yields a separate residual certificate for remaining endpoint motion. Exact parametric LP regions describe cutoff sensitivity across active-set changes. A feasible-repair theorem transfers uniform nonlinear constraint residual and objective-error bounds into contraction under a quantitative growth condition. Explicit examples show an exact rate and the failure of extrapolation from observed ratios. These results use established duality, order, and contraction arguments, with rational reference checks and explicit scope. A limited SCIP propagator implements numerical bound validation, current-round witness filtering, pilot stopping, and event-triggered reconsideration. The report distinguishes its comparative experiment from implementation of the stronger future-round certificate machinery. In a prospective 120-run comparison, native, fixed, and adaptive policies each solve the same 14 of 40 model–seed pairs. Adaptive uses 20.9% fewer directional LPs than fixed but 18.4% more measured sidecar time, with no additional solves. Native SCIP remains the default. An enhancement budget controls optional work on the modified search trajectory; it does not guarantee a runtime bound relative to an independent baseline. Certificate validity, cheap certificate discovery, and overall solver performance remain distinct claims.

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1 The decision problem and the scope of completion

Optimization-based bound tightening (OBBT) solves auxiliary optimization problems to improve variable bounds. The improvements can strengthen a nonlinear relaxation, simplify propagation, and reduce the search tree. They also cost time. A smaller tree is useful only if the saved work exceeds the cost of obtaining it. The question studied here is therefore when to start tightening, when to stop, and when a change in the search state makes it worth considering again.

The repository’s earlier study developed local contraction and stalling theory and tested iterated OBBT as external preprocessing. Its adaptive rules already stopped when observed contraction became weak. They did not produce a net general solver benefit: paying for the first round and restarting the solver could outweigh later reductions. Those outcomes are retained in [the September experiment report](#). This continuation does not relabel an observed contraction ratio as a new stopping theorem. It develops finite certificates for a specified tightening operator and evaluates a limited policy inside the search.

Three questions must remain separate. First, can a mathematical certificate bound the possible reduction of a coordinate? Second, can a particular implementation cheaply obtain such a certificate at a useful search state? Third, does acting on this information reduce total solving time? A proof answering the first question does not answer the other two. The same distinction applies to the word “complete.” This report completes a specified research program by giving explicit mathematical contracts, proofs, implementations, comparative evidence, and limitations. It does not resolve every open question about allocation of solver effort.

The central distinction is between a witness feasible in the current relaxation and a witness that remains available after relaxation rebuilding. The former bounds the benefit of a frozen round. The latter can protect an inner box throughout all subsequent rounds under precise monotonicity and cutoff assumptions. This difference determines what a solver may safely infer from a failed attempt to tighten a bound.

The report is self-contained at the level of the proved statements. Section 2 specifies the operator and its scope. Section 3 proves the finite certificates and their reuse conditions. Section 4 treats constrained contraction and its assumptions. Section 5 separates enhancement cost guarantees from guarantees relative to an independent baseline. Sections 6 and 7 describe the actual software and outcomes. The related-work account in Section 8 distinguishes classical ingredients from the present synthesis. Section 9 records the remaining frontier without treating an unsuccessful performance hypothesis as an implementation defect that must be hidden.

2 The tightening operator

Consider

$$f^* = \min\{f(x) : x \in X \cap B_0\}, \quad (1)$$

where $B_0 = [\ell^0, u^0] \subset \mathbb{R}^n$ is a bounded box and X includes the original constraints and any integrality restrictions. A relaxation built on $B \subseteq B_0$ has a projected objective $\phi_B : B \rightarrow \mathbb{R} \cup \{+\infty\}$: for a lifted formulation, $\phi_B(x)$ is the infimum of the relaxed objective over lifted points above x , with value $+\infty$ if none exist. Assume

$$\phi_B(x) \leq f(x) \quad (x \in X \cap B), \quad (2)$$

$$\phi_{B'}(x) \geq \phi_B(x) \quad (B' \subseteq B, x \in B'). \quad (3)$$

The second condition is a hypothesis on the whole relaxation construction, including any rows retained or discarded. It is not a consequence of calling a relaxation “McCormick” if unrelated strengthening rows change between calls. Assume that the nonempty projected sublevel sets used below are compact. This holds for the bounded lifted linear relaxations in the exact reference implementation.

For a finite objective cutoff U , define

$$K_U(B) = \{x \in B : \phi_B(x) \leq U\}, \quad T_U(B) = \text{box } K_U(B), \quad (4)$$

where box is the smallest axis-aligned closed box containing its argument. The empty set maps to the empty set. Exact Jacobi OBBT computes the coordinate minima and maxima on one frozen relaxation, then rebuilds the relaxation on the resulting box. A sequential variant may rebuild after each coordinate update; this difference matters for rate equalities.

Lemma 2.1 (Containment and order). *Every feasible point in $X \cap B$ with objective at most U belongs to $T_U(B) \subseteq B$. If $B' \subseteq B$, then $T_U(B') \subseteq T_U(B)$. If $U' \leq U$, then $T_{U'}(B) \subseteq T_U(B)$.*

Proof. Validity puts the stated feasible points in $K_U(B)$. The box hull stays in B . For $x \in K_U(B')$, monotonicity gives $\phi_B(x) \leq \phi_{B'}(x) \leq U$, so $K_U(B') \subseteq K_U(B)$; taking box hulls preserves inclusion. The cutoff assertion follows directly from sublevel-set inclusion. $\square$

If a feasible incumbent supplies $U \geq f^*$, exact iterates $B_{k+1} = T_U(B_k)$ are nested and preserve every global minimizer in B_0 . Their intersection B_∞ is a nonempty box. This elementary convergence statement does not supply a rate or a computable distance to B_∞ . Nor does nestedness alone justify identifying the intersection as a fixed point without an additional continuity property of the relaxation construction. The certificates below avoid that inference.

Write $w(B) = \max_i(u_i - \ell_i)$. A frequently useful bound from the earlier study illustrates the role of the incumbent and of curvature.

Proposition 2.2 (Quadratic growth and a relaxation error). *Suppose a global minimizer x^* and constants $\gamma > 0$, $\tau \geq 0$ satisfy, for all relevant B containing x^* and all $x \in B$,*

$$\phi_B(x) \geq f^* + \frac{\gamma}{2} \|x - x^*\|_\infty^2 - \tau w(B)^2.$$

With $\epsilon = U - f^* \geq 0$,

$$w(T_U(B))^2 \leq \frac{8\epsilon}{\gamma} + \frac{8\tau}{\gamma} w(B)^2. \quad (5)$$

If $q^2 = 8\tau/\gamma < 1$, then

$$w(B_k)^2 \leq q^{2k} w(B_0)^2 + \frac{8\epsilon}{\gamma} \frac{1 - q^{2k}}{1 - q^2}.$$

Proof. Every point of $K_U(B)$ is within infinity-norm distance $\sqrt{2(\epsilon + \tau w(B)^2)/\gamma}$ of x^* . Its coordinate ranges are at most twice this radius. Squaring gives (5); iteration of the affine scalar recurrence gives the second bound. $\square$

This is a sufficient contraction estimate, not a claim of an exact rate. It also explains why improving the incumbent may justify reconsidering tightening: the displayed floor scales with $\sqrt{U - f^*}$. In ordinary search, f^* , x^* , and a certified growth constant are not generally available. The estimate is therefore not itself a computable universal activation rule. Section 4 treats constrained support problems and makes the additional sensitivity assumptions explicit.

Finally, relaxation witnesses need not be feasible for the original nonlinear problem. A witness can prove that a relaxation cannot tighten a coordinate even when it violates a nonlinear equality or integrality. Conversely, a feasible incumbent justifies the cutoff but says little about the remaining width of a weak relaxation. Both types of information are useful, and they have different proof obligations.

3 Finite certificates for remaining benefit

3.1 Primal witnesses and a frozen round

Let $B = [\ell, u]$ and write $T_U(B) = [\ell', u']$. A point in the projected set $K_U(B)$ is a primal witness. In a lifted implementation, one must also supply and check its lifted coordinates.

Proposition 3.1 (Current-round ceilings). *Let W be a nonempty finite subset of $K_U(B)$. Then*

$$0 \leq \ell'_i - \ell_i \leq \min_{w \in W} w_i - \ell_i, \quad 0 \leq u_i - u'_i \leq u_i - \max_{w \in W} w_i. \quad (6)$$

In particular, a witness attaining ℓ_i or u_i proves that the corresponding directional bound cannot improve in this frozen round.

Proof. The minimum over all of $K_U(B)$ is no larger than the minimum over its subset W ; the maximum is no smaller than the maximum over W . Containment in B supplies nonnegativity. $\square$

This is the mathematical basis of established feasibility filtering. A relaxation optimum or a point returned for one coordinate may screen several other directions. Screening from an already available point can be cheap. Solving all $2n$ directional problems to obtain the witnesses can cost as much as the round being screened; a certificate alone provides no computational saving.

The quantifier “this frozen round” is essential. A witness may violate the relaxation rebuilt on a smaller box, even if its original coordinates remain in that box. Consequently, the current-round

ceiling is not an upper bound on total future tightening. For example, relax $w = x^2$ on $[0, u]$, $u > 0$, with square endpoint tangents and the secant, and impose $w \leq 0$. The upper support is $u/2$, attained at $(x, w) = (u/2, 0)$. That point violates the new upper-endpoint tangent after rebuilding on $[0, u/2]$. Repeated rounds halve the upper bound and converge to zero, so the eventual reduction is u , even though the first witness bounds the present reduction by $u/2$.

3.2 An inner box protected under all future rounds

Definition 3.2 (Protected box). A nonempty box $P = [a, b] \subseteq B$ is protected for the specified relaxation family and cutoff U if there is a finite witness pool $W \subseteq K_U(P)$ whose coordinate hull is P .

Feasibility is checked on the relaxation rebuilt on P , not merely on the current outer box B . Up to $2n$ witnesses suffice: one attaining each lower and upper face. Coincident faces can share witnesses.

Theorem 3.3 (Finite protection certificate). *Under (3), a protected box P satisfies $T_U(P) = P$. Every exact OBBT iterate starting at a box $B_0 \supseteq P$ contains P . More generally, this remains true for rounds with cutoffs $U_k \geq U$ and for any sequence of exact directional OBBT updates using the same monotone relaxation construction. Thus, for any such future iterate $B_k = [\ell^k, u^k]$ starting at $B = [\ell, u]$,*

$$0 \leq \ell_i^k - \ell_i \leq a_i - \ell_i, \quad 0 \leq u_i - u_i^k \leq u_i - b_i. \quad (7)$$

Proof. Since $W \subseteq K_U(P) \subseteq P$ and $\text{box } W = P$, taking coordinate hulls gives $T_U(P) = P$. If $P \subseteq B_k$ and $U_k \geq U$, then Lemma 2.1 gives $P = T_U(P) \subseteq T_{U_k}(B_k)$, proving the round assertion by induction. For a single directional update, the witness attaining the corresponding face of P remains feasible on the larger current box, so the new bound cannot cross that face. Induction over individual updates proves the sequential assertion. The endpoint inequalities are exactly $P \subseteq B_k \subseteq B$. $\square$

The same pool bounds the objective of the specified relaxation. Suppose its lifted points $\hat{w}$ use the same linear lifted objective $c^T \hat{w} + c_0$ and the lifted relaxation sets themselves are nested under box restriction, as in the reference family below. Set

$$U_W = \max_{\hat{w} \in W} (c^T \hat{w} + c_0), \quad V_W = \min_{\hat{w} \in W} (c^T \hat{w} + c_0). \quad (8)$$

The protection certificate remains reusable at any cutoff at least U_W , provided the current box contains P and the model and relaxation rows are unchanged. The minimum relaxed objective on every such future box is at most V_W , because the witnesses remain available. If L is a valid lower bound for that relaxed minimum, $V_W - L$ is an upper bound on its possible increase from L . It does not bound the improvement from cuts, integrality reasoning, or a different relaxation.

A smaller incumbent can cross U_W and invalidate this certificate. This gives an exact event at which reconsideration is justified, without claiming that paying for another round will be profitable. Domain changes can invalidate containment $P \subseteq B$, and additional cuts can invalidate witness feasibility. Both require a new check. A protected box for an LP relaxation may be removed by integer rounding; the theorem does not include that stronger operator unless its witnesses satisfy the corresponding premises. One sufficient extension is integral protected endpoints on every integer coordinate: if an exact lower bound is at most an integer face a_i , its ceiling is still at most a_i , and the analogous floor argument holds at b_i . This proves preservation under that rounding operation, but not under arbitrary native integer propagation.

The hull of a finite nonempty set of original feasible sublevel points is also protected whenever their exact lifts are available: the points attain every hull face, and relaxation validity supplies feasible lifts on the hull. Unlike generic relaxation witnesses, these original points survive every additional cut valid for that sublevel set. Producing exact feasible points for nonlinear equalities may itself be difficult.

3.3 Adapting witnesses to a new cutoff

In a fixed convex lifted relaxation, let v be a feasible witness and z another feasible point with objective $c(z) < c(v)$, where c includes the objective constant. For $c(z) \leq U' < c(v)$, define

$$\theta = \frac{U' - c(z)}{c(v) - c(z)}, \quad v' = z + \theta(v - z). \quad (9)$$

Convexity gives feasibility, and linearity gives $c(v') = U'$. This is an inexpensive way to produce current-round witnesses at a tighter cutoff. The point z is a relaxation anchor; it need not be feasible in the original nonlinear model. To certify a new protected box, one must still take the hull of the new points and check them again in the relaxation rebuilt on that hull. Convex mixing in an outer relaxation does not perform that last step.

Proposition 3.4 (Exact cutoff frontier of a finite witness pool). *Let $W = \{z_1, \dots, z_k\}$ be feasible points of one convex lifted relaxation with affine objective $c(z)$. Every coordinate minimum and maximum over $\text{conv}(W) \cap \{z : c(z) \leq U\}$ is attained either at an original point with $c(z_j) \leq U$ or at a mixture of a pair whose objectives lie strictly on opposite sides of U , chosen to have objective exactly U , provided that cutoff intersection is nonempty.*

Proof. Use convex coefficients $\alpha_j \geq 0$ with $\sum_j \alpha_j = 1$ and $\sum_j \alpha_j c(z_j) \leq U$. Every extreme coefficient vector either has one positive component or has two positive components and an active cutoff. Otherwise a nonzero perturbation preserving the one or two active affine equations keeps both signs feasible, contradicting extremality. Linear coordinate objectives attain extrema at these coefficient vectors. With two positive components the strict straddling gives the stated mixture; if both objective values equal U , a single original point suffices. $\square$

The `cutoff_frontier` routine generates these points using at most quadratically many pair mixtures and no new LP. It optimizes over the cached convex hull, not the full relaxation. An empty result says only that this pool cannot supply a witness. On a changed node box, the cached points must first pass the rebuilt row checks.

3.4 A residual certificate under a uniform contraction bound

An alternative certificate bounds the accumulated future motion of the OBBT endpoints. Represent a box by $p = (-\ell, u) \in \mathbb{R}^{2n}$, so shrinking a box decreases p componentwise, and let $F(p)$ represent one exact round at a fixed cutoff. The next result requires a uniform bound on a region, rather than a ratio observed on one trajectory.

Theorem 3.5 (A posteriori remaining motion). *Let $\mathcal{D}$ be a convex set of valid box-endpoint vectors that is invariant under F , and assume $F(p) \leq p$ on $\mathcal{D}$. Suppose a nonnegative matrix M satisfies, componentwise,*

$$|F(p) - F(q)| \leq M|p - q| \quad (p, q \in \mathcal{D}). \quad (10)$$

For $p_0 \in \mathcal{D}$, put $p_{k+1} = F(p_k)$ and $d = p_0 - p_1 \geq 0$. If a finite vector $e \geq 0$ satisfies

$$d + Me \leq e, \quad (11)$$

then the limit p_∞ exists and

$$0 \leq p_0 - p_\infty \leq e, \quad 0 \leq p_1 - p_\infty \leq e - d.$$

If $\rho(M) < 1$, one admissible choice is $e = (I - M)^{-1}d$.

Proof. Set $\Delta_k = p_k - p_{k+1} \geq 0$. The uniform estimate gives $\Delta_{k+1} \leq M\Delta_k$, so $\Delta_k \leq M^k d$. For each finite r , induction using (11) yields $\sum_{k=0}^r M^k d \leq e$: the base follows from $d \leq e$, and the next partial sum is at most $d + Me \leq e$. Therefore $0 \leq p_0 - p_{r+1} \leq e$. The decreasing endpoint sequence is bounded below, hence converges coordinatewise; taking limits proves the first bound. Subtracting the first movement d proves the second. If $\rho(M) < 1$, the nonnegative convergent Neumann series gives $e = \sum_{k \geq 0} M^k d$, which satisfies (11) with equality. $\square$

For a scalar contraction constant $q < 1$ in a compatible norm, the corresponding estimate is the familiar geometric-series bound $\|p_0 - p_\infty\| \leq \|p_0 - p_1\|/(1 - q)$. An observed ratio of two successive endpoint changes is not a proof of (10). Active-set changes can make the next round behave differently. A derivative bound at one point is likewise insufficient; its bound must cover the invariant region and every relevant active regime. The reference implementation checks the rational supersolution inequality. That arithmetic check does not manufacture the problem-specific uniform matrix bound.

The residual must be from an exact Jacobi round. A selective, interrupted, or inaccurate round may show zero movement while unprocessed directions still admit large changes. If a certified upper bound $\bar{d} \geq d$ is available, using $\bar{d} + Me \leq e$ still bounds total motion by e ; the after-round bound is $e - d$ when the exact first movement is known, not $e - \bar{d}$.

A finite rebuilt-McCormick example supplies its own uniform derivative bound. With $x \in [0, u]$, square endpoint tangents and secant, and cutoff $w \leq r^2$ for $u \geq r > 0$, the upper endpoint map is

$$F(u) = \frac{u^2 + r^2}{2u}, \quad F'(u) = \frac{1 - r^2/u^2}{2}.$$

The upper-endpoint tangent proves the upper bound, and the lift $(F(u), r^2)$ attains it. The interval $[r, u_0]$ is invariant, with uniform constant $q = (1 - r^2/u_0^2)/2 < 1/2$. For $r = 1, u_0 = 2$, the first residual is $3/4$ and $q = 3/8$, giving total-motion ceiling $6/5$; the actual total motion is 1. This bound covers the changing endpoint tangent coefficients, rather than assuming they are fixed.

If a common invariant region and cutoff interval admit the stronger uniform estimate

$$|F_U(p) - F_V(q)| \leq M|p - q| + h|U - V|, \quad h \geq 0, \quad \rho(M) < 1,$$

then any fixed points in that region satisfy $|p_U - p_V| \leq (I - M)^{-1}h|U - V|$. Substitute the fixed points, iterate the nonnegative inequality, and use $M^k \rightarrow 0$. Without this extra uniform estimate, an old residual certificate cannot be transferred to a tighter cutoff merely because the cutoff improved.

3.5 An exact reference checker

The reference family has real original variables x with rational box endpoints, auxiliary variables w_{ij} for selected products, fixed rational linear rows in (x, w) , and a rational linear objective. On $B = [\ell, u]$, each product uses the four inequalities

$$\begin{aligned} w_{ij} &\geq \ell_j x_i + \ell_i x_j - \ell_i \ell_j, & w_{ij} &\geq u_j x_i + u_i x_j - u_i u_j, \\ w_{ij} &\leq \ell_j x_i + u_i x_j - u_i \ell_j, & w_{ij} &\leq u_j x_i + \ell_i x_j - \ell_i u_j. \end{aligned}$$

The case $i = j$ is allowed. It then gives a finite linear relaxation of x_i^2 , with endpoint tangents below and a secant above, rather than the exact convex epigraph. Box restrictions and the linear objective cutoff complete the rows.

These product relaxations are monotone under box restriction. For distinct indices they describe the convex hull of the bilinear graph on the rectangle, whose hull decreases when the rectangle shrinks. For a square, each newly raised lower endpoint produces a tangent at least as high as the old lower-endpoint tangent on the smaller interval; the same holds at the upper endpoint, and the new secant is lower. Keeping the fixed rows therefore preserves the required inclusion.

`theory/certificates.py` represents coefficients and witnesses as exact rational numbers and checks every row. A floating-point LP may propose a witness, but only rational row checks issue a certificate. The checked candidate hull must be contained in the declared outer box, and its witnesses must attain every face. Failure to produce or rationally validate a witness returns an unresolved outcome; it is not evidence of absence of a protected box.

The certificate binds the full fixed model and relaxation construction. Equality of stored row arrays does not prove where a local row is valid. The caller must justify every fixed row on the original sublevel set and domain concerned before interpreting the LP certificate as a valid domain-reduction statement for that problem. This small checker establishes the witness implications above. It does not certify native SCIP rows, prove correctness of an entire nonlinear search, or convert a numerical incumbent into an exact feasible point. The operational policy evaluated in Section 7 is described separately in Section 6.

A ceiling on coordinate or relaxation-objective improvement is also limited to those measures. An OBBT call with zero immediate bound gain can still expose useful dual information or a generalized variable bound; Gleixner et al. discuss this under degeneracy in their Remark 5 [9]. A stopping certificate for endpoint movement therefore does not certify that every side effect of a new auxiliary solve is worthless.

4 Constraints, active-set changes, and cutoff reuse

Constraints can remove the directions responsible for a stall, but they also create active-set changes that invalidate a local extrapolation. Parametric LP duality gives precise contracts for some of this behavior. These are uses of classical sensitivity analysis; the important point here is the different conclusions supported by primal witnesses, dual vectors, and complete basis regions.

4.1 A fixed matrix and a changing right-hand side

Consider the family

$$R(\theta) = \{z : Az \leq b + E\theta\}, \quad h_c(\theta) = \max\{c^T z : z \in R(\theta)\}. \quad (12)$$

All parameters considered have nonempty relaxations and finite attained support values. The original variables are selected coordinates of z ; signed coordinate objectives $c = \pm e_i$ give OBBT endpoints. Exact certificates use rational data. Validity as an outer relaxation of the original nonlinear model is a separate premise.

The matrix A is fixed. Rebuilding McCormick rows generally changes their coefficients, so (12) does not automatically describe box changes. It applies directly to cutoff changes with a frozen box. Varying boxes require a valid normalization or separate verification of the coefficient dependence. A numerical basis is only a proposal for the checks below.

Proposition 4.1 (Dual envelope). *If $y \geq 0$ and $A^T y = c$, then*

$$h_c(\theta) \leq y^T b + y^T E \theta. \quad (13)$$

The minimum of finitely many such affine expressions is also an upper bound, even when their original bases cease to be optimal or feasible. If only the cutoff row $a^T z \leq U$ changes, a primal-dual pair is exactly optimal at U_0 , and $\eta \geq 0$ is its cutoff multiplier, then

$$h_c(U) \leq h_c(U_0) - \eta(U_0 - U) \quad (U \leq U_0). \quad (14)$$

Proof. For feasible z , $c^T z = y^T A z \leq y^T (b + E \theta)$. Maximizing and then taking the minimum over stored vectors proves the first statement. For a single changing cutoff, the dual affine expression has slope η and agrees with $h_c(U_0)$ at the certified optimum, giving the second statement. $\square$

The quantity $\eta(U_0 - U)$ is a guaranteed support reduction, not an upper bound on possible tightening. A zero multiplier provides no such guarantee; it does not prove that a larger cutoff change is ineffective. Without exact optimality at U_0 , the dual upper envelope remains valid, but subtracting from an unverified numerical optimum does not establish the displayed reduction.

Proposition 4.2 (Exact basis region). *Let $d = \dim z$, and choose d linearly independent rows I . Define*

$$z_I(\theta) = A_I^{-1}(b_I + E_I \theta), \quad y_I = A_I^{-T} c, \quad y_j = 0 \ (j \notin I).$$

If $y_I \geq 0$, then on the closed rational polyhedron

$$C_I = \{\theta : A z_I(\theta) \leq b + E \theta\}$$

the support is exactly

$$h_c(\theta) = c^T A_I^{-1}(b_I + E_I \theta). \quad (15)$$

Proof. The point $z_I(\theta)$ is feasible on C_I . The vector y is dual feasible and supported on rows tight at that point, so the primal and dual objective values agree. Proposition 4.1 supplies the matching upper bound. $\square$

Zero multipliers and ties are allowed; singular proposed bases are rejected. Checking the affine residuals at every vertex of a proposed parameter polytope proves its containment in C_I . This does not prove that a collection of such polytopes covers all parameters of interest. For a scalar cutoff, substituting $z_I(U) = v + wU$ makes every residual an affine inequality in U ; their exact intersection is a validity interval. Beyond it, the stored dual expression is still an upper bound but need not equal the support.

For an exact example, minimize y subject to $0 \leq x \leq 3$, $0 \leq y \leq 4$, and

$$x - y \leq 1, \quad 4x - y \leq 7, \quad x \leq 5/2.$$

With cutoff $y \leq U$, $0 \leq U \leq 4$, the upper support of x is

$$h_x(U) = \min\{1 + U, 7/4 + U/4, 5/2\} = \begin{cases} 1 + U, & 0 \leq U \leq 1, \\ 7/4 + U/4, & 1 \leq U \leq 3, \\ 5/2, & 3 \leq U \leq 4. \end{cases} \quad (16)$$

Choosing $y = U$ and the displayed x gives a feasible point meeting each upper bound, proving the formula. At $U_0 = 4$ the cutoff multiplier is zero, yet moving to $U = 2$ reduces the support from $5/2$ to $9/4$. The witness $(5/2, 3)$ proves no reduction only down to cutoff 3. At $U_0 = 2$, extrapolating the old dual to $U = 1/2$ gives the safe upper bound $15/8$, while the actual support is $3/2$.

4.2 Combining retained primal and dual information

For a signed coordinate objective c_i , let p_i be the corresponding current box endpoint. Keep all rows except the cutoff fixed. From stored feasible relaxed points W_i , retain those satisfying the new cutoff and define, when that subset is nonempty,

$$L_i(U) = \max\{c_i^T z : z \in W_i, a^T z \leq U\}.$$

If $F_i(U)$ is the exact support and $D_i(U)$ is a stored dual envelope, then

$$L_i(U) \leq F_i(U) \leq \min\{p_i, D_i(U)\}, \quad 0 \leq p_i - F_i(U) \leq p_i - L_i(U). \quad (17)$$

The left bound follows from primal feasibility, the right from duality and the current box. A near-endpoint witness certifies little possible one-round improvement; a sufficiently low dual envelope certifies a useful available improvement. The interval left between them is a natural place for a budgeted probe.

An old support-attaining witness proves the endpoint unchanged until its objective value is excluded. Finding the lowest-objective witness among old optima can itself require another LP, whose cost must be charged. An excluded witness does not prove that the support moves: another witness may remain. Every changed local row requires rechecking, and screening a whole round requires information for every direction whose benefit is bounded.

4.3 A uniform bound across active-set changes

Proposition 4.3 (Complete basis coverage implies a comparison matrix). *Fix the cutoff. Let $\mathcal{D}$ be a convex endpoint region for which, for each support coordinate i , finitely many verified closed basis regions cover $\mathcal{D}$. Suppose on each region $F_i(p) = \alpha_i + g_i^T p$. If $M \geq 0$ and $|g_{ij}| \leq M_{ij}$ for every covered region, then*

$$|F(p) - F(q)| \leq M|p - q| \quad (p, q \in \mathcal{D}).$$

Proof. The segment from p to q stays in $\mathcal{D}$. Its intersections with the finitely many polyhedral regions partition it into finitely many intervals. On each, the derivative of coordinate i along the segment has magnitude at most $\sum_j M_{ij}|p_j - q_j|$. At boundaries, both basis formulas equal the same optimal support, so there is no jump. Integrating over the segment gives the claim. $\square$

This provides the uniform premise in Theorem 3.5. It does not prove invariance of $\mathcal{D}$ or existence of a finite majorant e . For the order interval between a protected fixed box and the current box, monotonicity supplies invariance. A derivative bound only on the present basis region is insufficient if later rounds leave that region.

A deliberately simple example verifies all these steps. For $x^2 \leq 0$, $0 \leq x \leq r$, retain only tangents to x^2 at $t_1 = r/2$ and $t_2 = r/8 + 3/4$. For positive tangent location t , its necessary inequality $2tx - t^2 \leq 0$ is $x \leq t/2$. Together with the box, the upper-support map on $0 \leq r \leq 8$ is

$$F(r) = \min\{r/4, r/16 + 3/8\}. \quad (18)$$

At $r = 0$ the box fixes $x = 0$, so no division by a zero tangent coefficient is needed. The two basis regions are $[0, 2]$ and $[2, 8]$, with slopes $1/4$ and $1/16$. They cover the invariant interval, giving the uniform bound $M = [1/4]$. The trajectory

$$8 \mapsto 7/8 \mapsto 7/32 \mapsto 7/128 \mapsto \dots$$

crosses a basis boundary. Its first width ratio $7/64$ is smaller than the subsequent ratio $1/4$. The example tests a sensitivity certificate; direct nonlinear propagation would solve the original problem at once.

4.4 An equality changes the exact contraction rate

Consider $f(x, y) = (x - y)^2$, expanded as $x^2 + y^2 - 2xy$, on $[-r, r]^2$. Keep the convex square terms exact and use the McCormick overestimator for xy . The relaxed objective is

$$\phi_r(x, y) = x^2 + y^2 - 2r^2 + 2r|x - y|.$$

Without further constraints the minimizers (r, r) and $(-r, -r)$ preserve every endpoint at cutoff zero, so OBBT stalls. Impose instead the exact equality $y = -x$. On it, $\phi_r(t, -t) = 2t^2 + 4r|t| - 2r^2$.

Proposition 4.4 (Constrained exact rate and cutoff floor). *For cutoff $U \geq 0$, a Jacobi round on the symmetric box returns radius*

$$G_U(r) = \min\{r, \sqrt{2r^2 + U/2} - r\}.$$

At $U = 0$, the exact linear rate is $\sqrt{2} - 1$. At $U > 0$, from any $r_0 \geq \sqrt{U}/2$, the radii decrease to exactly $\sqrt{U}/2$.

Proof. Solving $2|t|^2 + 4r|t| - 2r^2 \leq U$ gives the stated root, restricted by the current radius. Set $a = \sqrt{U}/2$. The root is less than r precisely for $r > a$, and at least a because $2r^2 + 2a^2 - (r + a)^2 = (r - a)^2 \geq 0$. Hence the radii decrease to a limit at least a ; continuity and the fixed-point equation force that limit to be a . Factoring r gives the zero-cutoff rate. $\square$

Eliminating the equality before relaxing gives the objective $4x^2$. Its exact convex cutoff reaches $|x| \leq \sqrt{U}/2$ in one round. Thus the rate depends on the formulation and relaxation, not only on the original constrained problem. Affine equalities can also restrict the earlier tangent support problems to a nullspace when the original expansion has no first-order objective term and is uniform on the admitted shapes. General nonlinear active constraints additionally need uniform residual and remainder control. A first-order tangent cone alone does not establish a finite contraction theorem.

4.5 Nonlinear constraints through a feasible repair

A uniform constraint error bound provides a route that does not require guessing the next active set. Let S be the original feasible set and $x^* \in S$ a minimizer with value f^* . Consider boxes containing x^* in a fixed neighborhood, and a nonnegative constraint residual $v(x)$ that vanishes on S . Suppose every point x in each projected relaxation satisfies

$$\phi_B(x) \geq f(x) - aw(B)^2, \quad v(x) \leq bw(B)^2, \quad (19)$$

with uniform $a, b \geq 0$. Assume it has a feasible repair $y \in S$ such that, for uniform $\kappa, L \geq 0$, $0 < \theta \leq 1$, and $\mu > 0$,

$$\|x - y\|_\infty \leq \kappa v(x)^\theta, \quad (20)$$

$$f(y) \leq f(x) + L\|x - y\|_\infty, \quad (21)$$

$$f(y) \geq f^* + \mu\|y - x^*\|_\infty^2. \quad (22)$$

The repair need not stay in B , but must stay in the neighborhood where the last two estimates hold. These are uniform assumptions on all relaxed points, including infeasible ones, not consequences of a numerical KKT residual. The argument is a standard transfer of error and growth bounds; an antecedent in the repository is [the error-bound transfer note](#).

Theorem 4.5 (Constrained width bound from a feasible repair). *Under (19)–(22), with $U = f^* + \epsilon$, $\epsilon \geq 0$, set $r = \kappa(bw(B)^2)^\theta$. Then*

$$w(T_U(B)) \leq 2r + 2\sqrt{\frac{\epsilon + aw(B)^2 + Lr}{\mu}}. \quad (23)$$

For $\theta = 1$, let $K = \kappa b$ and $q = 2\sqrt{(a + LK)/\mu}$. If $q < 1$ and $\lambda = q + 2K\bar{w} < 1$ for some $\bar{w} > 0$, every such box with $w(B) \leq \bar{w}$ satisfies

$$w(T_U(B)) \leq \lambda w(B) + 2\sqrt{\epsilon/\mu}. \quad (24)$$

For nested iterates in the stated neighborhood,

$$\limsup_k w(B_k) \leq \frac{2\sqrt{\epsilon/\mu}}{1 - \lambda}.$$

At zero cutoff slack ($\epsilon = 0$), the widths contract with upper ratio λ . If $K = 0$, take $\lambda = q$ on any admitted width neighborhood.

Proof. For a retained relaxed point, $f(x) \leq f^* + \epsilon + aw(B)^2$. Its repair has distance at most r and objective at most $f^* + \epsilon + aw(B)^2 + Lr$. Feasible growth bounds its distance to x^* by the square root in (23). The triangle inequality bounds the distance of x by this radius plus r . Taking coordinate ranges doubles the bound. If $\theta = 1$, split the square root using $\sqrt{s+t} \leq \sqrt{s} + \sqrt{t}$ and bound $2Kw(B)^2 \leq 2K\bar{w}w(B)$. Nesting keeps all later widths at most $\bar{w}$, and summing the scalar geometric recurrence gives the stated limit bound. $\square$

The limit bound is an upper bound, not an exact floor. At zero cutoff slack, with $\theta < 1$ and fixed positive $L\kappa b$, the square-root term in (23) has order w^θ , so this argument need not establish linear contraction. This is a limitation of the estimate, not proof of a stall. A local repair-pair constant L_B of order $w^{2-2\theta}$ can recover an $O(w)$ square-root term, but the separate repair term $O(w^{2\theta})$ also needs control.

An explicit nonlinear equality illustrates the uniform assumptions. Let

$$S = \{(x, y) : y = x^2\} \cap ([-1/4, 1/4] \times [0, 1/16]), \quad f(x, y) = x^2 + y^2 - xy/16.$$

The minimizer is $(0, 0)$ with value zero. For a subbox $B = [\ell, u] \times [0, v]$ containing the origin, use

$$x^2 \leq y \leq (\ell + u)x - \ell u, \quad \phi_B(x, y) = x^2 + y^2 - \frac{1}{16} \min\{uy, \ell y + vx - \ell v\}.$$

The square epigraph and secant relax the graph, and the displayed bilinear overestimator gives a convex objective estimator. A relaxed point has feasible repair (x, x^2) in the same box, with distance $y - x^2 \leq (u - \ell)^2/4 \leq w(B)^2/4$. The bilinear-envelope error is at most $(u - \ell)v/4$, so one may take

$$a = 1/64, \quad b = 1/4, \quad \kappa = 1, \quad \theta = 1.$$

On the outer box, $|\partial_x f| + |\partial_y f| \leq (2 + 1/16)(1/4 + 1/16) = 165/256$, so take $L = 165/256$. On the feasible graph,

$$f(x, x^2) = x^2(1 - x/16 + x^2) \geq (63/64)x^2 = (63/64)\|(x, x^2)\|_\infty^2.$$

Thus $\mu = 63/64$, $K = 1/4$, and $q^2 = 181/252 < 49/64$. For $w(B) \leq 1/8$, the rational upper rate $\lambda = 15/16$ gives

$$w(T_U(B)) \leq \frac{15}{16}w(B) + 2\sqrt{64\epsilon/63}.$$

This covers every admitted nested box shape without choosing an active set. Its constants are conservative. The example uses exact convex square epigraphs, which the production finite-tangent LP does not supply; it proves a constrained theorem, not the rate of the measured plugin.

4.6 Finite observed histories do not certify an eventual stall

Fix $N \geq 2$, and set $a = 1/2$, $\delta = 1/(2N)$, $b = a - \delta$, and $q = 1/4$. On $[0, 1]$ define

$$F_{\text{stall}}(s) = \min\{s, \max\{b, s - \delta\}\},$$

and

$$F_{\text{shrink}}(s) = \begin{cases} qs, & s \leq b, \\ qb + (b - qb)(s - b)/(a - b), & b \leq s \leq a, \\ s - \delta, & s \geq a. \end{cases}$$

Both maps are continuous and nondecreasing, with $0 \leq F(s) \leq s$; these assertions follow from the nonnegative slopes and the endpoints of each affine piece. Thus $R_s = [0, F(s)]$ is a valid monotone outer relaxation family for the feasible set $\{0\}$. Starting at $s_0 = 1$, both trajectories subtract δ through $s_{N+1} = b$. One then stalls at b , and the other converges geometrically to zero. By increasing N , arbitrarily many common early ratios can be made close to one despite the different eventual outcomes.

These are not asserted to be standard McCormick relaxations. They show that validity, monotonicity, and a finite observed history alone cannot certify the remaining benefit. A protected box, a uniform sensitivity bound, or further relaxation structure supplies information absent from that history. Exact checks of the examples and basis procedures appear in [theory/check_constrained_obbt.py](#).

5 What a cost budget can guarantee

The allocation results in this section are elementary accounting and scheduling facts. They place useful limits on the performance claims one can derive from a guarded enhancement. No publication priority is claimed for them.

Proposition 5.1 (Cost on the enhanced execution). *Let N be the time charged to native work on one enhanced execution and E the time charged to optional enhancement work. Admit a new enhancement only when $E < b + \eta N$, where $b, \eta \geq 0$. If each admitted operation adds at most δ to E , and every increment of E belongs to such an admitted operation, then at every completed operation*

$$E \leq b + \eta N + \delta, \quad N + E \leq (1 + \eta)N + b + \delta. \quad (25)$$

If each cost has an enforced upper bound r and admission instead requires $E + r \leq b + \eta N$, the final overshoot δ can be removed.

Proof. Immediately before the last admitted operation, the admission condition holds. That operation adds at most δ ; later native work only increases the right-hand side. If no operation was admitted, the result is immediate. An advance reservation that includes the complete eventual cost prevents the final overshoot. $\square$

The charge must include discovery, model assembly, screening, optimization, verification, and callback bookkeeping. A time limit on an auxiliary LP does not bound surrounding Python or symbolic work. An implementation without an enforced bound on the whole operation must record overshoot rather than claim that (25) holds with its nominal LP limit. If rejected entry or admission checks incur an unreserved cost J , that cost must be added separately to the total. The proposition gives no bound on J without an additional premise limiting it.

The quantity N refers to native work along the enhanced trajectory. Let N_0 denote the work in an independent baseline run. Nothing in Proposition 5.1 connects N and N_0 . An abstract example makes the missing premise clear. At the first decision a deterministic search procedure has two legal choices. Its default ordering finds a complete proof in one unit. Adding a redundant row changes the ordering and leads to a proof after M units. The row costs $b > 0$ to obtain. The enhancement budget is respected, but the runtime ratio is $M + b$, which is unbounded as M grows. This is a counterexample in an abstract search model, not a constructed SCIP instance or a prediction of SCIP behavior.

An independent, unchanged baseline provides a different guarantee.

Proposition 5.2 (Protected baseline scheduling). *In an ideal preemptible work model, run independent solvers A and H with separate states and fixed random tapes. Suppose by total work time t , solver A has received at least $(1 - \eta)t - h$ work and H at least $\eta t - h$, for $0 < \eta < 1$ and $h \geq 0$. If their standalone completion requirements are T_A, T_H , the first answer arrives by*

$$\min \left\{ \frac{T_A + h}{1 - \eta}, \frac{T_H + h}{\eta} \right\}. \quad (26)$$

Alternatively, run H for at most b charged work, then, if necessary, start unchanged A from its initial state. The total work is at most $b + T_A$, provided startup is charged.

Proof. At either time in (26), the corresponding solver has received its required work. For serial rescue, either H finishes within b , or its cost at most b is followed by exactly the independent baseline execution. $\square$

The assumptions exclude interference from shared memory exhaustion, cache contention, uncharged startup, and wall-clock-dependent decisions. Injecting a new incumbent or rows into the baseline can change its work requirement and needs a separate argument. This report does not deploy a solver portfolio or claim the scheduling bound for its experiments.

There is also no universally correct first action from an arbitrary reward history. Suppose two actions each cost one unit and two possible environments have identical observable histories. Only the first action reveals a proof in one environment; only the second does so in the other. A deterministic selector fails to choose correctly first in one environment. A randomized first choice succeeds with probability at most one half in at least one environment. Structural information or a statistical assumption can distinguish the cases; history alone cannot. This observation does not rule out learning or competitive scheduling.

These limits suggest a small interface for optional solver actions. Each action declares its model, node domain, relaxation, cutoff, and cost allowance. It returns a verified inference, a verified upper bound on benefit in a stated mathematical measure, or an unresolved status. A selector may rank unresolved actions heuristically, but must not turn a ranking into a validity claim. The ledger charges selection as well as optimization. An OBBT width reduction, a support-cut violation, and a decomposition lower-bound improvement are different measures; converting them to expected seconds saved requires a separately validated model. The present implementation and comparison concern the OBBT case.

6 Implementation and numerical scope

The software has two separate roles. The exact reference modules check the mathematical certificate formats on declared rational relaxation families. The integrated solver prototype applies a limited scheduling policy to a numerical search. Its outcomes test that policy, not the automatic discovery of protected boxes or uniform contraction matrices.

6.1 Exact reference modules

`theory/certificates.py` checks current-round witness ceilings, protected boxes, cutoff mixing, objective ceilings, and residual supersolutions. Exact-data constructors reject floating-point inputs: intended coefficients must be integers, rational strings, or rational numbers. The optional SciPy proposal routine can solve up to $2n$ endpoint LPs, rationalize the proposed points, and check their hull. It returns no certificate if any exact row check fails. An optional additional objective LP proposes a stronger objective ceiling after a protected box is found; only its primal feasibility must be verified. Its cost is also charged. This complete check of a proposed certificate is not a complete search for every certificate that might exist.

`theory/check_constrained_obbt.py` verifies exact basis calculations and the explicit changing-cell and cutoff examples. It does not enumerate basis regions of an arbitrary solver model. The remaining-motion majorant routine checks $d + Me \leq e$; the caller is responsible for the uniform sensitivity and invariance premises. The distinction prevents a successful arithmetic test from being mistaken for an unsupported theorem about the native solver’s complete OBBT map.

6.2 A standalone certified stopping workflow

`theory/certified_driver.py` makes the protected-box check operational for the rational reference family. A numerical solver proposes a primal point z and multipliers α for each endpoint LP $\min c^T z$ subject to $Az \leq b$. The driver reconstructs rationals and checks exactly

$$Az \leq b, \quad \alpha \leq 0, \quad A^T \alpha = c, \quad c^T z = \alpha^T b.$$

Weak duality proves the endpoint optimum. All variable bounds are explicit rows. The driver commits a Jacobi round only after every endpoint proof passes, then rechecks its witness pool on the rebuilt coordinate hull. If that check succeeds, it stops with a protected fixed box. An optional objective proposal needs only verified primal feasibility to improve the objective ceiling.

The statuses distinguish a certified fixed box, an exhausted round budget (**unfinished**), and a failed exact reconstruction (**inconclusive**). An incomplete round is discarded while previous certified rounds remain valid. Exact termination need not occur in finite time, and rational reconstruction can fail even when the underlying mathematical LP is well behaved. The driver retains proof objects and reports all LP calls and elapsed time including exact checks. This is a complete workflow for checking and using the stated stopping certificate; it is not a complete exact LP solver or an all-future stopping rule for native SCIP’s changing relaxation.

6.3 A limited policy inside SCIP

`solver/adaptive_obbt.py` builds the same original quadratic constraints in SCIP for all arms. It normalizes maximization to minimization and uses an epigraph variable for a quadratic objective. The additional propagator uses current local bounds of the transformed original variables to construct a separate linear relaxation. Native SCIP OBBT remains enabled in every arm. The campaign uses SCIP 10.0.2 through PySCIPOpt 6.2.1, with Python 3.13.11 and SciPy 1.18.1. The newer pinned SCIP source inspected for the literature audit is a separate artifact, not the benchmark binary.

The side relaxation contains the original linearized quadratic rows, McCormick inequalities for bilinear products, square tangents and a secant, auxiliary interval bounds, and an incumbent cutoff when available. The square relaxation uses five current tangent locations and a bounded pool of retained tangents. It is a finite LP approximation, not the exact square epigraph used in Proposition 4.4. It also differs from the endpoint-only square relaxation of the smallest exact

reference checker. The theoretical statements apply to their declared families; their quantitative rates are not transferred between them.

The fixed and adaptive arms share the same total directional LP allowance. The adaptive arm orders nonlinear-variable directions using quadratic coefficient and current-width scores, rechecks cached witnesses to screen directions with small possible current-round gain, and stops a pilot after unproductive probes. Scheduling state is indexed by node. A first visit, a sufficiently improved incumbent, or a sufficiently reduced box can admit another call, subject to depth, call, LP, and time limits. Cutoff and box-change thresholds are policy parameters, not certified predictions of runtime savings.

The frozen default allowance is 64 directional LPs in total, at most 12 per root call and 4 per nonroot call, 24 calls in total, 3 calls per node, and depth at most 64. Each auxiliary LP has a nominal 0.25-second limit, and sidecar work has a nominal allowance of 10% of the run's time budget. The adaptive pilot uses two LPs and stops if the accumulated relative tightening is below 10^{-3} . A 1% scaled cutoff improvement or a 10% decrease in a coordinate width can retrigger a call. The campaign source snapshot is the authority for every parameter actually used in the experiments.

Cached primal points are rechecked against the current frozen side LP before screening. A direction is screened when a checked point is within $10^{-7} \max\{1, u_i - \ell_i\}$ of its endpoint. This is a tolerance ceiling for the frozen LP, not proof that every future round is useless. The protected-box and matrix-tail certificates are not used to control the measured policy. No claim is made that a cheaply available all-future-round certificate was demonstrated on the application cohort.

6.4 Validated sidecar bounds

The auxiliary numerical LP solver proposes dual information. The propagator does not use a reported primal coordinate optimum directly as a lower bound. The following residual formula explains its validation.

Proposition 6.1 (A bound from any signed dual vector). *Consider minimizing $c^T z$ over $Az \leq b$ and a finite box $z \in [\ell, u]$. For any $y \leq 0$, set $r = c - A^T y$. Then*

$$\min c^T z \geq y^T b + \sum_j \min\{r_j \ell_j, r_j u_j\}. \quad (27)$$

No exact dual stationarity is required.

Proof. Since $y \leq 0$ and $Az \leq b$, one has $y^T Az \geq y^T b$. Therefore $c^T z = y^T Az + r^T z \geq y^T b + r^T z$. Minimizing the latter linear residual independently over the finite coordinate intervals gives (27). $\square$

The implementation rounds the products, sums, and residual intervals outward, rejecting nonfinite inputs or overflow. This validates a bound for the stored binary floating-point LP rows. The row builder separately forms exact dyadic expressions from the imported binary data and rounds the coefficients outward. For an exact row $a^T z \leq \beta$, a rounded coefficient vector $\tilde{a}$ is made valid by choosing

$$\tilde{\beta} \geq \beta + \sum_j \max\{(\tilde{a}_j - a_j)\ell_j, (\tilde{a}_j - a_j)u_j\}.$$

Every point satisfying the exact row on the box then satisfies the rounded row. The rounding arithmetic and residual-bound tests concern this imported binary model, not an independent reconstruction of exact decimal or symbolic data upstream of the importer.

The plugin adds a further outward margin before submitting a local bound to SCIP. It rejects a proposal outside the current domain and checks compatibility with integer rounding. It never turns

a side-LP infeasibility report into a node cutoff. Unsupported finite-domain or arithmetic cases are recorded and leave native search available. A cached witness from another node is only a proposal; passing every current row check is required before use.

These safeguards do not constitute a certificate for the entire SCIP solve. In particular, the cutoff comes from a numerically accepted SCIP incumbent after a numerical original-model residual check. Its objective is evaluated exactly for the imported binary coefficients and incumbent coordinates, rounded upward, and given a safety margin. Exact objective evaluation does not prove exact original-model feasibility. The conditional validity of cutoff-derived reductions still depends on that premise. Native presolve, propagation, branching, and optimality claims are not replayed by an independent checker. Returned solutions receive the declared numerical original-model validation in the experiment, and this is reported as numerical validation.

6.5 Cost and failure accounting

The run deadline includes model construction and optimization. Once the initial admission checks pass, the sidecar timer charges box inspection, construction, witness checks, LP solves, validation, and application time. The earlier budget, depth, and node checks are outside that timer; their cost remains in complete process elapsed time. Expensive setup can overrun the nominal sidecar allowance before the next admission check. An LP time limit is likewise subject to the numerical solver’s return behavior. LP construction checks its deadline every 64 generated rows, and cached witness screening also checks the callback deadline. The implementation records these costs; it does not claim the exact ledger bound in Proposition 5.1 with a hard whole-callback value of δ .

The experiment measures complete worker process elapsed time separately, including startup, parsing, validation, and output. Exceptions, unsupported models, failed LP proposals, missing incumbents, and time-limit overshoot remain visible. A failure to produce a bound has an unresolved status; it is not a certificate of no benefit or infeasibility.

7 Prospective comparison and verification

7.1 Exact certificate demonstrations

The exact checks and LP proposals are recorded in [the certificate-check results](#) and [the certified-driver results](#). Automatic discovery finds protected boxes on six of eight fixed fixtures: three simple square-stall examples, a three-variable quadratic with nonnegative square variables, and two linear models. On both strictly contracting square fixtures the proposals do not certify a fixed box. Those outcomes are inconclusive, not evidence that useful tightening is absent. Every accepted certificate is replayed exactly.

The three-variable example minimizes

$$f(x) = \sum_{i=1}^3 x_i^2 + \sum_{i<j} x_i x_j = \frac{1}{2} \left(\|x\|_2^2 + (x_1 + x_2 + x_3)^2 \right)$$

on $[-1, 1]^3$, with unique optimum zero at the origin. The lifted LP has square variables $s_i \geq 0$, McCormick product variables w_{ij} , and cutoff $\sum_i s_i + \sum_{i<j} w_{ij} \leq 0$. At each face $x_i = \pm 1$, set the other original coordinates to zero, $s_i = 1$, other squares to zero, the products involving i to zero, and the opposite product to -1 . These feasible lifted points protect all six faces. At the origin, squares zero and all three products -1 give objective -3 , so every later OBBT relaxation has lower bound at most -3 . Nonnegative squares and $w_{ij} \geq -1$ prove that the current LP bound is exactly

–3. The witness also satisfies exact convex square epigraphs: negative square variables do not cause this stall. This instantiates the earlier study’s row-stall mechanism.

Table 1 records representative standalone driver outcomes. The linear coupled example reaches its fixed box after one exact round. The zero-cutoff square contracts by one half each round and correctly remains unfinished after five rounds. For the positive-cutoff square, three rounds are certified; a later rational reconstruction fails an exact primal row check. The last completed box is retained, with an inconclusive status. This is an implementation precision limit, not a mathematical stall.

Fixture	Rounds	LPs	Status	Retained box or upper bound
Quadratic stall	1	7	fixed	$[-1, 1]^3$
Linear coupling	1	4	fixed	$[1/2, 2]^2$
Negative linear interval	1	2	fixed	$[-3, -1]$
Zero-cutoff square	5	10	unfinished	$u = 1/32$
Positive-cutoff square	3	6	unfinished	$u = 3281/3280$
Same, further round attempted	3	8	inconclusive	$u = 3281/3280$

Table 1: Exact reference workflow. The quadratic-stall LP count includes one objective proposal. All other shown calls are endpoint proposals. Recorded times include exact checks; these tiny fixtures test correctness, not solver performance.

Additional checks cover zero round budgets, invalid dual proofs, rollback of incomplete rounds, preservation of earlier certified rounds, invalid optional objective proposals, model and domain changes, degenerate boxes, cutoff mixing, cached-pool frontiers, and the residual majorant. The constrained checker verifies basis regions, the changing-cell example, and the exact constants and identities in the nonlinear feasible-repair example. These targeted checks supplement the proofs; they do not replace their uniform assumptions.

7.2 Frozen protocol

The protocol in [experiments/PROTOCOL.md](#) selects 12 public models and 8 synthetic diagnostics before the current comparative outcomes. Public admission uses the earlier September study’s metadata: between 5 and 300 variables, between 1 and 200 nonlinear variables, at most 500 quadratic terms, and at least five seconds in every available historical native SCIP baseline run. Eligible names are sorted by the SHA-256 digest of a fixed study prefix and the name; the first twelve are retained, at most one per historical family. The entire eligibility list and input hashes are frozen. This is a selected cohort informed by separate historical results, not a random sample of MINLP applications.

The synthetic structures are point packing, coupled square budgets, bilinear cycles, and sparse indefinite quadratic programs, with two sizes of each and generator seed 20261003. Distinct tiny pilot instances exercise APIs and callback coverage and are excluded from the comparison. Synthetic outcomes are reported separately from public-instance outcomes.

Each model is run with SCIP seed shifts 0 and 1 in native, fixed, and adaptive arms, for 120 planned jobs. The native arm has no additional propagator. The fixed and adaptive arms share the LP allowance and native settings; the adaptive arm adds ordering, witness screening, and pilot stopping. Native SCIP OBBT stays enabled throughout. No known optimum, external feasible solution, or reference warm start enters the experiment. The task order is shuffled with seed 20261003. Numerical libraries and SCIP use one thread, with at most two worker processes at a time.

Each solver call has a ten-second budget including model construction and sidecar work. Process elapsed time additionally includes startup, parsing, validation, and output and is recorded separately. Every returned incumbent is evaluated against the original bounds, integrality, quadratic constraints, and objective. The declared numerical acceptance threshold is 10^{-6} for the maximum scaled constraint residual and integrality residual. An optimal status with an invalid incumbent counts as unsuccessful. This is a floating-point diagnostic, not exact feasibility certification.

The primary reported penalized time uses full process elapsed time for successful jobs and 20 seconds for unsuccessful jobs (PAR-2). The one-second-shift geometric mean reported below uses full process time. Archived capped means instead use $\min\{t, 10\}$ for process time t ; they are diagnostic, can hide overruns, and do not penalize a fast unsuccessful termination. Raw total process time exposes startup and overrun costs. The normalized final gap is

$$\min \left\{ 1, \frac{\max\{0, p - d\}}{\max\{1, |p|, |d|\}} \right\},$$

where p is a valid numerical incumbent objective and d is a finite dual bound; missing admissible values receive score one. Paired gap comparisons report their eligible denominators. No timing statistic is restricted silently to cases solved by every arm. The two seeds are repeated runs on each model, not forty independent application families.

7.3 Complete campaign outcomes

All 120 scheduled jobs in `campaign-01` completed. Each arm solved the same 14 of 40 model-seed pairs: 2 of 24 public pairs and 12 of 16 synthetic pairs. Neither extra policy added or lost a solve. Each arm had one no-incumbent run: `genpooling_meyer04`, seed zero. No process failure, invalid incumbent, material objective mismatch, or inconsistent primal-dual pair was reported under the declared numerical checks.

Cohort	Arm	Solved	Mean PAR-2	Shifted process mean	Mean gap score
All	Native	14/40	13.855	5.957	0.20351
All	Fixed	14/40	13.988	6.285	0.18438
All	Adaptive	14/40	13.997	6.265	0.18806
Public	Native	2/24	18.964	10.248	0.33408
Public	Fixed	2/24	18.999	10.297	0.30209
Public	Adaptive	2/24	19.026	10.324	0.30798
Synthetic	Native	12/16	6.192	2.384	0.00767
Synthetic	Fixed	12/16	6.471	2.772	0.00782
Synthetic	Adaptive	12/16	6.453	2.733	0.00818

Table 2: Prospective outcomes. Times are seconds. PAR-2 includes every run; the geometric mean uses full observed process time with a one-second shift and no cap. Lower gap scores are better, with missing valid incumbent or finite consistent dual bound scored as one.

Summed full process elapsed times were 308.050 seconds for native, 313.407 for fixed, and 313.618 for adaptive. The corresponding uncapped one-second-shift geometric means were 5.957, 6.285, and 6.265 seconds. These totals retain startup, validation, output, and deadline overrun. The largest solver-call elapsed time was 10.093 seconds; the largest complete process time was 10.606 seconds. The experiments shared a 36-logical-CPU machine with unrelated studies; starting load averages were 17.76, 15.65, and 14.11. The small timing differences do not support a claim of a stable speed advantage for either extra policy. The comparison provides no evidence of a net solving benefit.

The adaptive policy used 1,719 directional LPs compared with 2,172 for fixed, a 20.9% reduction. This did not reduce measured sidecar time: adaptive used 23.123 seconds versus fixed’s 19.530 seconds, an 18.4% increase. Adaptive entered 741 calls, including 639 nonroot calls and 43 incumbent retriggers; fixed entered 404 calls, including 300 nonroot calls and 39 incumbent retriggers. Accepted bound changes numbered 308 and 112, respectively. These counts show that the experiment exercised in-tree tightening and retriggers. The lower LP count did not translate into lower complete plugin cost; the additional calls also required construction and checks.

The adaptive policy screened only two directions from cached feasible witnesses. It stopped 574 calls after an unproductive pilot, 113 at the per-call LP allowance, and six at a time budget. The corresponding fixed counts were 349 LP-budget stops and seven time-budget stops. Both extra arms recorded 48 unsupported-domain calls, all on `qp3`, where the current box was unbounded for this sidecar. These calls remain in the costs and outcomes. Four fixed and 26 adaptive directional attempts did not supply a usable bound; individual LP status details for those attempts were not retained, so their causes are not inferred.

Mean public gap scores improved relative to native, with much of the change concentrated in `bayes2_50` under both seeds. Among 23 public pairs with finite comparable gaps, adaptive improved four and worsened ten by more than 10^{-4} ; fixed improved five and worsened ten. Across all 39 comparable pairs, adaptive had four gap wins and twelve losses. The averages reflect the sizes of changes as well as their counts. Across all 40 pairs, adaptive had two PAR-2 wins and eleven losses exceeding 5% relative to native; fixed had one win and ten losses. These are descriptive counts from a small, short-budget cohort, not statistical evidence across independent problem families. The practical conclusion is to retain native SCIP as the default and keep the additional policy experimental.

7.4 Provenance and targeted checks

The campaign directory `experiments/runs/campaign-01` contains immutable input and source hashes, the actual worker source, every raw outcome and solver log, per-run analysis, and the complete summary. The policy and analyzer were frozen before holdout outcomes. During the uninterrupted campaign, review found that an interrupted attempt could be overwritten by the live runner’s resume path. The live runner was fixed to archive incomplete attempts before retrying. This recovery-only change did not alter the running campaign, its frozen sources, or any measured attempt; a separate targeted recovery check passed. The frozen campaign remains the authority for its reported data.

The final combined solver, reviewer, and experiment suite passed 31 targeted tests. Separate exact checks cover the certificate contracts, constrained-theory examples, and optional-work accounting. A direct OSiL evaluator checked all twelve normalized public models independently of the experiment importer. The completed campaign audit checked all 120 expected jobs and numerically validated all 69 returned public incumbents against the original OSiL data. The document build checks references, citations, and PDF metadata. These are local checks for this continuation; no project-wide verification or CI status/log inspection is represented as evidence here.

The exact commands and outcomes are recorded in `VERIFICATION.md` and the `document README`. The `claim and evidence map` distinguishes proved statements, checked proposals, numerical validation, and empirical outcomes. Internal independent review is not formal verification or external peer review.

8 Related work and attribution

Selective OBBT, learned variable selection, reuse after incumbent improvements, and adaptive scheduling of solver components all have substantial prior work. The relevant gap is narrower than the phrase “adaptive OBBT” suggests.

Gleixner, Berthold, Müller, and Weltge develop primal filtering, LP ordering, and Lagrangian variable bounds [9]. Their methods are direct antecedents of screening and cheap dual reuse. The inspected SCIP implementation already includes feasibility filtering, aggregate filtering, budgets, and generalized variable bounds. Its separate generalized-bound propagator detects improved cutoffs and reevaluates stored bounds [16]. Coramin implements feasible-solution and aggregate-objective filtering and attributes these techniques to the same OBBT literature [8]. Therefore, incumbent awareness or skipping some coordinate LPs cannot carry a broad novelty claim.

Cengil et al. dynamically select OBBT variables using learned rankings for AC optimal power flow [6]. Their experiments address relaxation strengthening for phase-angle differences; the result is not a general MINLP runtime guarantee. Gómez-Casares et al. study conic OBBT, dual-information propagation, and instance-level learned configuration selection in polynomial optimization [10]. Application-specific work also studies the balance between bound quality and computation for ReLU networks and selective integrality in tightening subproblems [1, 15]. Explicitly retaining redundant bounds on lifted variables can improve or worsen complete solver performance [11]. These findings support charging full cost and measuring complete solves.

The mathematical ingredients here are established. Nested fixed sets and monotone iterations belong to order-theoretic fixed-point reasoning and prior domain-reduction analysis [17, 4, 5, 2]. The residual estimate is a matrix geometric-series argument of the Banach–Perov type [14]. Exact basis regions are standard multiparametric LP objects [3]. The report supplies their OBBT-specific premises, finite witnesses, cutoff and domain reuse rules, and checked examples. It does not propose a new general order theorem, contraction principle, or LP duality theorem.

The protected-box construction differs from ordinary current-round filtering in one essential check: its witnesses must be feasible in the relaxation rebuilt on their own coordinate hull. That establishes the persistence needed for an all-future-round claim. Likewise, a residual certificate needs a uniform sensitivity bound over a verified invariant region, including every active-set change. A sequence of observed contraction ratios is not such a bound. These precise contracts, and their distinction from the implemented heuristic policy, are the contribution assessed here. This targeted literature audit does not establish publication priority for every formulation of those contracts.

Broader computation selection also has precedents. SCIP studies use bandits for expensive components and schedule diving and large-neighborhood heuristics jointly [13, 7]. Hay et al. formalize the value of further computation and explicit stopping in a probabilistic decision model [12]. Applying those ideas here requires assumptions on how strengthening changes the later search. Anderson acceleration can propose fixed-point candidates [18]; unconstrained mixing does not make such candidates valid domain reductions. A proposed protected box can be checked independently, but it is a lower limit on tightening, not permission to delete the rest of the current search domain.

The source ledger in [literature/source-ledger.md](#) records the sources and pinned software revisions inspected on October 3, 2026. The accompanying [related-work account](#) states the audit limits. Absence from these selected sources is not evidence that a subject has never been studied.

9 What is established and what remains open

The developed mathematical chain specifies the benefit being bounded, the relaxation to which the statement applies, the event that invalidates reuse, and the exact arithmetic needed to verify a proposed certificate. It handles current-round screening, all-future-round protection, conditional residual bounds, cutoff sensitivity, and active-set changes under their respective assumptions. The implementations and exact examples make those statements reviewable. The solver experiment provides no demonstrated net solving benefit for the tested additional policy; native SCIP remains the default. Fewer directional LPs did not mean lower complete sidecar cost.

Several stronger interpretations do not follow. A protected inner box does not authorize shrinking the search domain to that box: it limits what the specified OBBT procedure can remove. An exact bound on endpoint motion does not bound useful dual side effects, discrete propagation, or total search time. A certificate for fixed LP rows need not survive new cuts, integer rounding, or a different relaxation. A numerical solve with validated sidecar bounds is not an independently certified complete MINLP solve. A cap on optional work does not preserve the baseline’s search trajectory or runtime.

The remaining research questions are concrete. Cheap discovery of protected boxes from points already generated by a native solver may make future-round certificates operationally useful; the existence of the checker alone does not demonstrate that saving. Uniform sensitivity certificates for rebuilt nonlinear relaxations need bounds on changing coefficients and coverage of all reachable regimes. Applying the nonlinear repair theorem to broader model classes requires usable uniform residual, repair, growth, and objective-error constants. A controller choosing among cuts, OBBT, decomposition, branching, and incumbent search needs a validated model connecting their different mathematical benefits to later search work.

These are genuine open extensions, not missing claims silently supplied by the current software. The finite-history example and the enhanced-tree runtime counterexample identify why stronger conclusions need additional information. They do not prove that every useful restricted version is impossible. Completion of this report means completion of its stated proofs, implementations, checks, and experiment; it does not mean that adaptive solver effort allocation is an exhausted research area.

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